

Table1

```
-- Employees Table
CREATE TABLE employees (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    dept_id INT,
    salary INT,
    hire_date DATE,
    city VARCHAR(50)
);

INSERT INTO employees VALUES
(101, 'Anuj', 1, 50000, '2022-01-10', 'Delhi'),
(102, 'Ravi', 2, 60000, '2021-03-15', 'Mumbai'),
(103, 'Neha', 1, 55000, '2020-07-20', 'Delhi'),
(104, 'Amit', 3, 70000, '2019-11-01', 'Pune'),
(105, 'Sneha', 2, 65000, '2023-05-12', 'Mumbai'),
(106, 'Karan', 3, 72000, '2018-08-25', 'Bangalore'),
(107, 'Pooja', 1, 48000, '2022-09-30', 'Delhi'),
(108, 'Raj', 2, 52000, '2021-12-11', 'Chennai'),
(109, 'Meena', 3, 68000, '2020-02-14', 'Pune'),
(110, 'Vikas', 1, 51000, '2023-01-18', 'Delhi');
```

```
mysql> SELECT * FROM employees;
```

```
+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | dept_id | salary | hire_date | city |
+-----+-----+-----+-----+-----+-----+
| 101 | Anuj | 1 | 50000 | 2022-01-10 | Delhi |
| 102 | Ravi | 2 | 60000 | 2021-03-15 | Mumbai |
| 103 | Neha | 1 | 55000 | 2020-07-20 | Delhi |
| 104 | Amit | 3 | 70000 | 2019-11-01 | Pune |
| 105 | Sneha | 2 | 65000 | 2023-05-12 | Mumbai |
| 106 | Karan | 3 | 72000 | 2018-08-25 | Bangalore |
| 107 | Pooja | 1 | 48000 | 2022-09-30 | Delhi |
| 108 | Raj | 2 | 52000 | 2021-12-11 | Chennai |
| 109 | Meena | 3 | 68000 | 2020-02-14 | Pune |
| 110 | Vikas | 1 | 51000 | 2023-01-18 | Delhi |
+-----+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

Table 2

```
-- departments table
DROP TABLE IF EXISTS departments;
CREATE TABLE departments (
    dept_id INT PRIMARY KEY,
    dept_name VARCHAR(50)
);

INSERT INTO departments VALUES
(1, 'HR'),
(2, 'Sales'),
(3, 'IT');
```

```
mysql> SELECT * FROM departments;
```

```
+-----+-----+
| dept_id | dept_name |
+-----+-----+
|      1 | HR       |
|      2 | Sales    |
|      3 | IT       |
+-----+-----+
3 rows in set (0.00 sec)
```

Table3

```
-- Projects Table
DROP TABLE IF EXISTS projects;
CREATE TABLE projects (
    project_id INT PRIMARY KEY,
    project_name VARCHAR(50),
    dept_id INT,
    budget INT
);

INSERT INTO projects VALUES
(201, 'Alpha', 1, 100000),
(202, 'Beta', 2, 150000),
(203, 'Gamma', 3, 200000),
(204, 'Delta', 2, 120000);
```

```
mysql> SELECT * FROM projects;
```

```
+-----+-----+-----+-----+
| project_id | project_name | dept_id | budget |
+-----+-----+-----+-----+
|          201 | Alpha        |        1 | 100000 |
|          202 | Beta         |        2 | 150000 |
|          203 | Gamma        |        3 | 200000 |
|          204 | Delta        |        2 | 120000 |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

Table4

```
-- Attendance table
```

```
DROP TABLE IF EXISTS attendance;
```

```
CREATE TABLE attendance (
    emp_id INT,
    date DATE,
    status VARCHAR(20)
);
```

```
INSERT INTO attendance VALUES
```

```
(101, '2024-03-01', 'Present'),
(102, '2024-03-01', 'Absent'),
(103, '2024-03-01', 'Present'),
(104, '2024-03-01', 'Present'),
(105, '2024-03-01', 'Absent'),
(106, '2024-03-01', 'Present'),
(107, '2024-03-01', 'Present'),
(108, '2024-03-01', 'Present'),
(109, '2024-03-01', 'Absent'),
(110, '2024-03-01', 'Present');
```

```
mysql> SELECT * FROM attendance;
```

```
+-----+-----+-----+
| emp_id | date       | status |
+-----+-----+-----+
| 101 | 2024-03-01 | Present |
| 102 | 2024-03-01 | Absent  |
| 103 | 2024-03-01 | Present |
| 104 | 2024-03-01 | Present |
| 105 | 2024-03-01 | Absent  |
| 106 | 2024-03-01 | Present |
| 107 | 2024-03-01 | Present |
| 108 | 2024-03-01 | Present |
| 109 | 2024-03-01 | Absent  |
| 110 | 2024-03-01 | Present |
+-----+-----+-----+
10 rows in set (0.00 sec)
```

SQL Questions (Lab Practice)

Basic Level (1–10)

1. Retrieve all employees

```
SELECT * FROM employees;
```

```
+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | dept_id | salary | hire_date | city |
+-----+-----+-----+-----+-----+-----+
| 101 | Ask ChatGPT | 1 | 50000 | 2022-01-10 | Delhi |
| 102 | Ravi | 2 | 60000 | 2021-03-15 | Mumbai |
| 103 | Neha | 1 | 55000 | 2020-07-20 | Delhi |
| 104 | Amit | 3 | 70000 | 2019-11-01 | Pune |
| 105 | Sneha | 2 | 65000 | 2023-05-12 | Mumbai |
| 106 | Karan | 3 | 72000 | 2018-08-25 | Bangalore |
| 107 | Pooja | 1 | 48000 | 2022-09-30 | Delhi |
| 108 | Raj | 2 | 52000 | 2021-12-11 | Chennai |
| 109 | Meena | 3 | 68000 | 2020-02-14 | Pune |
| 110 | Vikas | 1 | 51000 | 2023-01-18 | Delhi |
+-----+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

2. Display only emp_name and salary.

```
SELECT emp name, salary FROM employees;
```

```
+-----+-----+
| emp_name | salary |
+-----+-----+
| Anuj      | 50000 |
| Ravi      | 60000 |
| Neha      | 55000 |
| Amit      | 70000 |
| Sneha     | 65000 |
| Karan     | 72000 |
| Pooja     | 48000 |
| Raj       | 52000 |
| Meena     | 68000 |
| Vikas     | 51000 |
+-----+-----+
10 rows in set (0.00 sec)
```

3. Find employees working in department 1.

```
SELECT * FROM employees WHERE dept_id = 1;
```

```
+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | dept_id | salary | hire_date | city |
+-----+-----+-----+-----+-----+-----+
99 Ask ChatGPT | 1 | 50000 | 2022-01-10 | Delhi |
| 103 | Neha | 1 | 55000 | 2020-07-20 | Delhi |
| 107 | Pooja | 1 | 48000 | 2022-09-30 | Delhi |
| 110 | Vikas | 1 | 51000 | 2023-01-18 | Delhi |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

4. List employees from Delhi

```
SELECT * FROM employees WHERE city = 'Delhi';
```

```
+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | dept_id | salary | hire_date | city |
+-----+-----+-----+-----+-----+-----+
| 101 | Anuj | 1 | 50000 | 2022-01-10 | Delhi |
| 103 | Neha | 1 | 55000 | 2020-07-20 | Delhi |
| 107 | Pooja | 1 | 48000 | 2022-09-30 | Delhi |
| 110 | Vikas | 1 | 51000 | 2023-01-18 | Delhi |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

5. Show employees with salary greater than 60000.

```
SELECT emp name, salary FROM employees WHERE salary > 60000;
```

```

+-----+-----+
| emp_name | salary |
+-----+-----+
| Amit      | 70000  |
| Sneha     | 65000  |
| Karan     | 72000  |
| Meena     | 68000  |
+-----+-----+
4 rows in set (0.00 sec)

```

6. Find employees hired after 2021.

```
SELECT * FROM employees WHERE hire date > '2021-01-01';
```

```

+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | dept_id | salary | hire_date | city  |
+-----+-----+-----+-----+-----+-----+
| 101    | Anuj     | 1       | 50000  | 2022-01-10 | Delhi |
| 105    | Sneha    | 2       | 65000  | 2023-05-12 | Mumbai |
| 107    | Pooja    | 1       | 48000  | 2022-09-30 | Delhi |
| 108    | Raj      | 2       | 52000  | 2021-12-11 | Chennai |
| 110    | Vikas    | 1       | 51000  | 2023-01-18 | Delhi |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

```

7. Display unique cities.

```
SELECT DISTINCT city FROM employees;
```

```

+-----+
| city  |
+-----+
| Delhi |
| Mumbai |
| Pune  |
| Bangalore |
| Chennai |
+-----+
5 rows in set (0.00 sec)

```

8. Sort employees by salary (descending).

```
SELECT emp name, salary FROM employees ORDER BY salary DESC;
```

```

+-----+-----+
| emp_name | salary |
+-----+-----+
| Karan    | 72000 |
| Amit     | 70000 |
| Meena    | 68000 |
| Sneha    | 65000 |
| Ravi     | 60000 |
| Neha     | 55000 |
| Raj      | 52000 |
| Vikas    | 51000 |
| Anuj     | 50000 |
| Pooja    | 48000 |
+-----+-----+
10 rows in set (0.00 sec)

```

9. Count total employees.

```
SELECT COUNT(*) FROM employees;
```

```

+-----+
| COUNT(*) |
+-----+
|        10 |
+-----+
1 row in set (0.00 sec)

```

10. Find average salary.

```
SELECT AVG(salary) FROM employees;
```

```

+-----+
| AVG(salary) |
+-----+
|        61100.0 |
+-----+
1 row in set (0.00 sec)

```

Intermediate Level (11–20)

11. Find total salary paid per department.

```
SELECT dept_id, SUM(salary) FROM employees GROUP BY dept_id;
```

```
+-----+-----+
| dept_id | SUM(salary) |
+-----+-----+
|      1 |      204000 |
|      2 |      177000 |
|      3 |      210000 |
+-----+-----+
3 rows in set (0.00 sec)
```

12. Count employees in each department.

```
SELECT dept_id, SUM(salary) FROM employees GROUP BY dept_id;
```

```
+-----+-----+
| dept_id | COUNT(*) |
+-----+-----+
|      1 |         4 |
|      2 |         3 |
|      3 |         3 |
+-----+-----+
3 rows in set (0.00 sec)
```

13. Find highest salary in each department.

```
SELECT dept_id, MAX(salary) FROM employees GROUP BY dept_id;
```

```
+-----+-----+
| dept_id | MAX(salary) |
+-----+-----+
|      1 |       55000 |
|      2 |       65000 |
|      3 |       72000 |
+-----+-----+
3 rows in set (0.00 sec)
```

14. Display employees along with department names.


```
SELECT e.emp_name, d.dept_name
FROM employees e
JOIN departments d
ON e.dept id = d.dept id;
```

```
+-----+-----+
| emp_name | dept_name |
+-----+-----+
| Anuj      | HR        |
| Ravi      | Sales     |
| Neha      | HR        |
| Amit      | IT        |
| Sneha     | Sales     |
| Karan     | IT        |
| Pooja     | HR        |
| Raj       | Sales     |
| Meena     | IT        |
| Vikas     | HR        |
+-----+-----+
10 rows in set (0.00 sec)
```

15. List employees working on Sales department.

```
SELECT e.emp_name FROM employees e JOIN departments d ON e.dept id = d.dept id WHERE d.dept name = 'Sales';
```

```
+-----+
| emp_name |
+-----+
| Ravi      |
| Sneha     |
| Raj       |
+-----+
3 rows in set (0.00 sec)
```

16. Find employees whose name starts with 'A'.

```
SELECT * FROM employees WHERE emp name LIKE 'A%';
```

```

+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | dept_id | salary | hire_date | city |
+-----+-----+-----+-----+-----+-----+
| 101 | Anuj | 1 | 50000 | 2022-01-10 | Delhi |
| 104 | Amit | 3 | 70000 | 2019-11-01 | Pune |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

```

17. Get employees with salary between 50000 and 70000.

```
SELECT * FROM employees WHERE salary BETWEEN 50000 AND 70000;
```

```

+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | dept_id | salary | hire_date | city |
+-----+-----+-----+-----+-----+-----+
| 101 | Anuj | 1 | 50000 | 2022-01-10 | Delhi |
| 102 | Ravi | 2 | 60000 | 2021-03-15 | Mumbai |
| 103 | Neha | 1 | 55000 | 2020-07-20 | Delhi |
| 104 | Amit | 3 | 70000 | 2019-11-01 | Pune |
| 105 | Sneha | 2 | 65000 | 2023-05-12 | Mumbai |
| 108 | Raj | 2 | 52000 | 2021-12-11 | Chennai |
| 109 | Meena | 3 | 68000 | 2020-02-14 | Pune |
| 110 | Vikas | 1 | 51000 | 2023-01-18 | Delhi |
+-----+-----+-----+-----+-----+-----+
8 rows in set (0.00 sec)

```

18. Find employees not from Mumbai.

```
SELECT * FROM employees WHERE city != 'Mumbai';
```

```

+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | dept_id | salary | hire_date | city |
+-----+-----+-----+-----+-----+-----+
| 101 | Anuj | 1 | 50000 | 2022-01-10 | Delhi |
| 103 | Neha | 1 | 55000 | 2020-07-20 | Delhi |
| 104 | Amit | 3 | 70000 | 2019-11-01 | Pune |
| 106 | Karan | 3 | 72000 | 2018-08-25 | Bangalore |
| 107 | Pooja | 1 | 48000 | 2022-09-30 | Delhi |
| 108 | Raj | 2 | 52000 | 2021-12-11 | Chennai |
| 109 | Meena | 3 | 68000 | 2020-02-14 | Pune |
| 110 | Vikas | 1 | 51000 | 2023-01-18 | Delhi |
+-----+-----+-----+-----+-----+-----+
8 rows in set (0.00 sec)

```

19. Show employees hired in 2022.

```
SELECT * FROM employees WHERE YEAR(hire date) = 2022;
```

```

+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | dept_id | salary | hire_date | city |
+-----+-----+-----+-----+-----+-----+
| 101 | Anuj | 1 | 50000 | 2022-01-10 | Delhi |
| 107 | Pooja | 1 | 48000 | 2022-09-30 | Delhi |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

```

20. Find top 3 highest paid employees.

```
SELECT * FROM employees ORDER BY salary DESC LIMIT 3;
```

```

+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | dept_id | salary | hire_date | city |
+-----+-----+-----+-----+-----+-----+
| 106 | Karan | 3 | 72000 | 2018-08-25 | Bangalore |
| 104 | Amit | 3 | 70000 | 2019-11-01 | Pune |
| 109 | Meena | 3 | 68000 | 2020-02-14 | Pune |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

```

Joins (21–25)

21. INNER JOIN Employees and Departments.

```

SELECT e.emp_name, d.dept_name
FROM employees e
JOIN departments d
ON e.dept_id = d.dept_id;

```

```

+-----+-----+
| emp_name | dept_name |
+-----+-----+
| Anuj      | HR        |
| Ravi      | Sales     |
| Neha      | HR        |
| Amit      | IT        |
| Sneha     | Sales     |
| Karan     | IT        |
| Pooja     | HR        |
| Raj       | Sales     |
| Meena     | IT        |
| Vikas     | HR        |
+-----+-----+
10 rows in set (0.00 sec)

```

22. LEFT JOIN to show all employees with department names.

```

SELECT e.emp_name, d.dept_name
FROM employees e
LEFT JOIN departments d
ON e.dept id = d.dept id;

```

```

+-----+-----+
| emp_name | dept_name |
+-----+-----+
| Anuj      | HR        |
| Ravi      | Sales     |
| Neha      | HR        |
| Amit      | IT        |
| Sneha     | Sales     |
| Karan     | IT        |
| Pooja     | HR        |
| Raj       | Sales     |
| Meena     | IT        |
| Vikas     | HR        |
+-----+-----+
10 rows in set (0.00 sec)

```

23. Find employees working in IT department.

```
SELECT e.emp_name
FROM employees e
JOIN departments d
ON e.dept_id = d.dept_id
WHERE d.dept_name = 'IT';
```

```
+-----+
| emp_name |
+-----+
| Amit     |
| Karan    |
| Meena    |
+-----+
3 rows in set (0.00 sec)
```

24. Show project names with department names.

```
SELECT p.project_name, d.dept_name
FROM projects p
JOIN departments d
ON p.dept_id = d.dept_id;
```

```
+-----+-----+
| project_name | dept_name |
+-----+-----+
| Alpha        | HR        |
| Beta         | Sales     |
| Gamma        | IT        |
| Delta        | Sales     |
+-----+-----+
4 rows in set (0.00 sec)
```

25. Find employees and their project budgets (via department).

```
SELECT e.emp_name, p.budget
FROM employees e
JOIN projects p
ON e.dept_id = p.dept_id;
```

```

+-----+-----+
| emp_name | budget |
+-----+-----+
| Anuj     | 100000 |
| Neha     | 100000 |
| Pooja    | 100000 |
| Vikas    | 100000 |
| Ravi     | 150000 |
| Sneha    | 150000 |
| Raj      | 150000 |
| Ravi     | 120000 |
| Sneha    | 120000 |
| Raj      | 120000 |
| Amit     | 200000 |
| Karan    | 200000 |
| Meena    | 200000 |
+-----+-----+
13 rows in set (0.00 sec)

```

Advanced Level (26–30)

26. Find second highest salary.

```

SELECT MAX(salary)
FROM employees
WHERE salary < (SELECT MAX(salary) FROM employees);

```

```

+-----+
| MAX(salary) |
+-----+
|          70000 |
+-----+
1 row in set (0.00 sec)

```

27. Rank employees by salary.

```
SELECT emp_name, salary,
RANK() OVER (ORDER BY salary DESC) AS rnk
FROM employees;
```

```
+-----+-----+-----+
| emp_name | salary | rnk |
+-----+-----+-----+
| Karan    | 72000  | 1   |
| Amit     | 70000  | 2   |
| Meena    | 68000  | 3   |
| Sneha    | 65000  | 4   |
| Ravi     | 60000  | 5   |
| Neha     | 55000  | 6   |
| Raj      | 52000  | 7   |
| Vikas    | 51000  | 8   |
| Anuj     | 50000  | 9   |
| Pooja    | 48000  | 10  |
+-----+-----+-----+
10 rows in set (0.00 sec)
```

28. Find employees earning above average salary.

```
SELECT *
FROM employees
WHERE salary > (SELECT AVG(salary) FROM employees);
```

```
+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | dept_id | salary | hire_date | city |
+-----+-----+-----+-----+-----+-----+
| 104    | Amit     | 3       | 70000  | 2019-11-01 | Pune |
| 105    | Sneha    | 2       | 65000  | 2023-05-12 | Mumbai |
| 106    | Karan    | 3       | 72000  | 2018-08-25 | Bangalore |
| 109    | Meena    | 3       | 68000  | 2020-02-14 | Pune |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

29. Get department with highest total salary.

```
SELECT dept_id, SUM(salary) AS total
FROM employees
GROUP BY dept_id
ORDER BY total DESC
LIMIT 1;
```

```
+-----+-----+
| dept_id | total |
+-----+-----+
|      3 | 210000 |
+-----+-----+
1 row in set (0.00 sec)
```

30. Find employees who were absent on 2024-03-01.

```
SELECT e.emp_name
FROM employees e
JOIN attendance a
ON e.emp_id = a.emp_id
WHERE a.status = 'Absent';
```

```
+-----+
| emp_name |
+-----+
| Ravi      |
| Sneha     |
| Meena     |
+-----+
3 rows in set (0.00 sec)
```